

CHAPTER 1

INTRODUCTION

1.1. Introduction

The rapid proliferation of deepfake technology—driven by advances in generative adversarial networks, autoencoders, and diffusion-based text-to-video models—has created an unprecedented threat to information integrity across journalism, legal proceedings, social media, and personal privacy. Modern deepfakes range from traditional face-swap and expression-reenactment manipulations to fully AI-generated videos produced by tools such as Sora, Pika, and CogVideo, making it increasingly difficult for both humans and automated systems to distinguish authentic media from fabricated content. Existing detection approaches predominantly rely on a single forensic cue, such as spatial artifacts or frequency-domain anomalies, which renders them inherently brittle against compression artifacts encountered on social media platforms and vulnerable to novel generation techniques that leave no trace in the specific signal being analyzed.

This project presents a Multi-Cue Deep Learning Framework for Deepfake Video Detection that addresses these challenges by combining six complementary forensic detection cues—visual, temporal, remote photoplethysmography (rPPG), frequency, audio-visual synchronization, and perceptual trajectory straightening (ReStraV)—into a unified, attention-based fusion architecture. Each cue independently analyzes a distinct dimension of video authenticity: spatial face artifacts, cross-frame temporal coherence, physiological plausibility, spectral manipulation fingerprints, lip-sync consistency, and AI-generated motion characteristics, respectively. The system is trained and evaluated on four large-scale benchmark datasets—Google DFD, FakeAVCeleb, FaceForensics++ (C23), and VidProM—encompassing over 97,000 videos across seven or more manipulation types and seven AI-video generators.

1.2. Objective

The objective of this project is to design and develop a multi-cue deep learning framework for deepfake video detection that combines six complementary forensic cues—visual, temporal, rPPG, frequency, audio-visual synchronization, and ReStraV—into an attention-based fusion model, capable of accurately detecting both traditional face-swap manipulations and fully AI-generated videos with robustness against real-world compression artifacts encountered on social media platforms.

The project further aims to achieve cross-dataset generalization by training and evaluating the framework across four diverse benchmark datasets—Google DFD, FakeAVCeleb, FaceForensics++, and VidProM—covering over seven manipulation techniques and seven AI-video generators. Additionally, the system incorporates quality-aware inference mechanisms, including blur detection, compression assessment, and face resolution gating, to ensure reliable performance under varying video conditions. A user-friendly web interface is also developed to enable real-time video analysis and provide an interpretable 1–5 rating verdict, making the framework practical and accessible for end-users seeking to verify the authenticity of video content.

1.3. Problem Statement

Deepfake videos—synthetically manipulated media generated using deep learning techniques such as generative adversarial networks, autoencoders, and diffusion-based models—have emerged as a serious threat to information integrity, public trust, and digital security. These manipulations range from face-swapping and facial reenactment to fully AI-generated videos produced by advanced tools like Sora, Pika, CogVideo, and ModelScope. The ease of creating highly realistic fake videos has raised critical concerns across domains including journalism, legal evidence authentication, political misinformation, and personal privacy. As deepfake generation methods continue to evolve rapidly, the need for reliable and robust automated detection systems has become more urgent than ever.

Existing deepfake detection approaches predominantly rely on a single forensic cue—such as spatial artifacts, frequency-domain anomalies, or temporal inconsistencies—to classify videos as real or fake. While these methods may perform well under controlled laboratory conditions, they are inherently brittle and fail to generalize when faced with novel manipulation techniques, heavy video compression typical of social media platforms (YouTube, Instagram, WhatsApp), or the emerging class of fully AI-generated videos that leave no traditional face-manipulation traces. There is, therefore, a critical need for a comprehensive, multi-perspective detection framework that can analyze videos across multiple forensic dimensions simultaneously and adapt to diverse manipulation types, compression levels, and emerging AI-generation threats.

CHAPTER 2

LITERATURE SURVEY

2.1. Explainable Multimodal Deepfake Detection[1]

Ijaz Ul Haq et al. (2024) proposed a multimodal neurosymbolic approach for explainable deepfake detection. Their method is based on the biological observation that human emotions are difficult to imitate consistently in manipulated videos. The framework integrates deep learning models with symbolic reasoning, using facial expressions, lip synchronization, and emotion-based features extracted through CNNs. This combination allows the system to provide interpretable detection decisions rather than black-box outputs.

While the approach improves robustness and explainability, the study reports limitations when handling extreme head poses, occlusions, and non-frontal facial views. Additionally, the method requires accurate facial recognition, which may not always be achievable in unconstrained real-world videos. This indicates that explainable multimodal systems, although promising, still face practical challenges.

2.2. Audiovisual Deepfake Detection: Techniques and Challenges [2]

Ammarah Hashmi et al. (2024) presented a comprehensive survey on audiovisual deepfake detection, analyzing state-of-the-art methods that combine audio and visual information. Their work reviews GAN-based generation techniques, variational autoencoders, transformer networks, and audio-visual synchronization analysis. The survey also considers human perception factors, highlighting how humans detect inconsistencies between speech and facial motion.

Although the survey provides a broad understanding of existing audiovisual detection techniques, it does not propose a new detection model. The authors emphasize the difficulty of standardizing human perception factors and the need for real-time, generalizable detection systems. This work highlights the gap between theoretical research and practical deployment of audiovisual deepfake detectors.

2.3. Multimodal Deep Learning Approaches for Audio-Visual Fusion

Luciano Lopez (2025) explored multimodal deep learning architectures for enhanced audio-visual deepfake detection. The proposed approach uses ResNet-based CNNs for visual feature extraction and MFCC-based spectral analysis for audio processing. Attention mechanisms and

transformer layers are employed to fuse audio and visual features effectively, improving temporal modeling and classification performance.

The study demonstrates that multimodal feature fusion enhances detection accuracy by leveraging complementary information from audio and video streams. However, it also introduces significant computational overhead and requires balanced datasets for effective training. These constraints limit the applicability of such models in resource-constrained or real-time environments.

2.4. Surveys on Deepfake Detection Across Multiple Modalities

Rami Mubarak et al. (2023) conducted a broad survey on deepfake detection across visual, audio, and textual formats. The study reviews CNN-based spatial feature extraction, LSTM-based temporal sequence modeling, and GAN-based discriminative approaches. It also discusses the broader societal impact of deepfakes in political, economic, and social contexts.

While the survey provides a holistic overview of detection methodologies, its wide scope limits detailed analysis of individual techniques. The authors emphasize the need for unified, adaptable, and real-time deepfake detection systems that can generalize across diverse datasets. This reinforces the importance of building modular and extensible detection frameworks.

2.5. Summary and Research Gap

From the reviewed literature, it is evident that single-cue deepfake detection methods are insufficient for real-world scenarios where deepfakes are generated using increasingly diverse and sophisticated techniques. While multimodal and explainable approaches have shown improved reliability, they continue to face significant challenges related to computational complexity, dataset imbalance, and limited generalization across unseen manipulation methods. These findings directly motivated the development of the present multi-cue detection framework. Rather than relying on any single modality, the system integrates five independent detection cues — visual, temporal, physiological, frequency-domain, and audio-visual — each designed to capture a distinct category of manipulation artifact. The cues are combined through an attention-based fusion model that learns to assign importance weights based on each cue's reliability. Additionally, the system is trained across multiple diverse datasets with balanced sampling, and evaluated using cross-dataset benchmarks to validate its generalization capability. This design addresses the core gaps identified in the reviewed literature: the need for modular extensibility, multi-dataset robustness, and interpretable decision-making in deepfake detection.

CHAPTER 3

BLOCK DIAGRAM

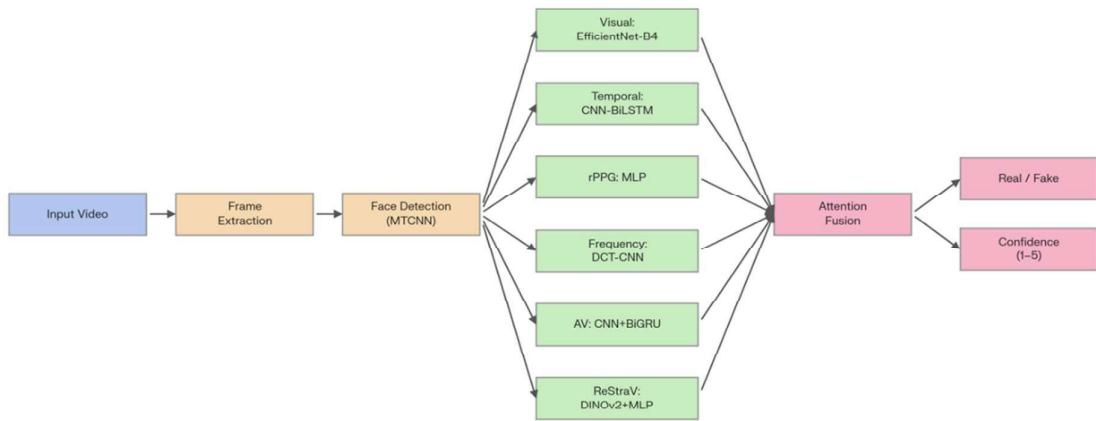


Fig 3.1. Block diagram

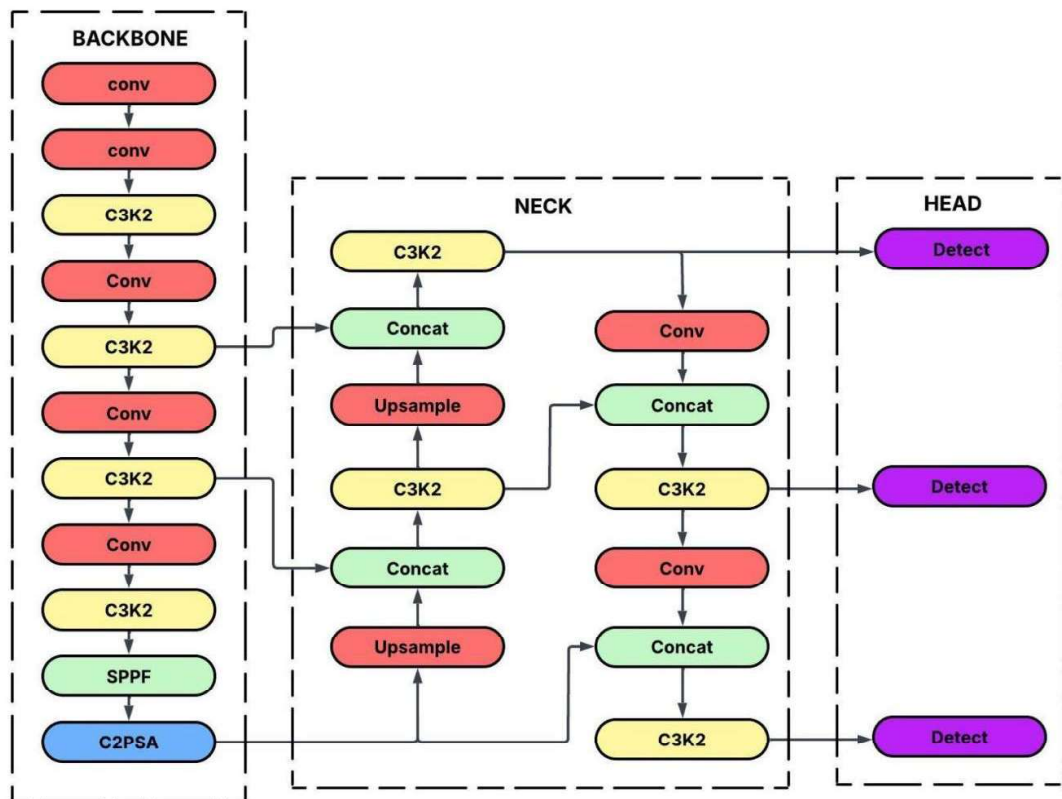


Fig 3.2. Block diagram of the YOLOv11 model

CHAPTER 4

FLOWCHART

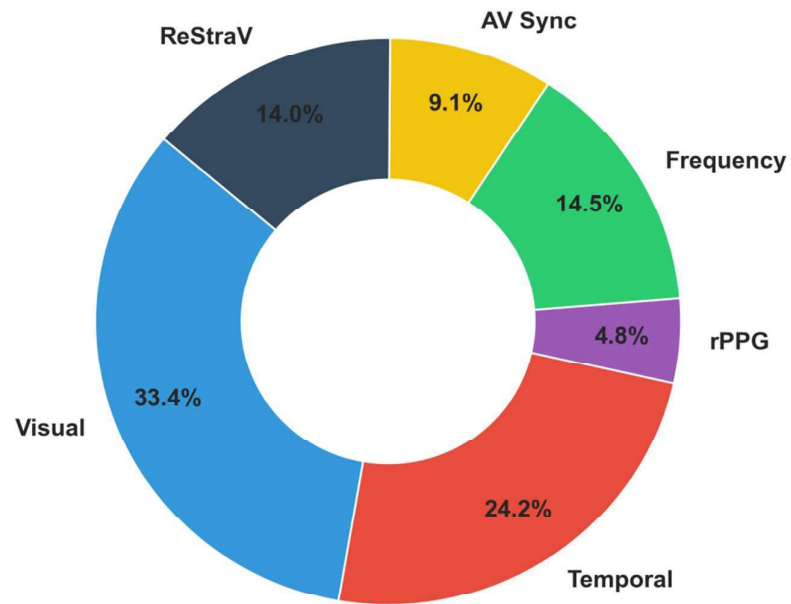
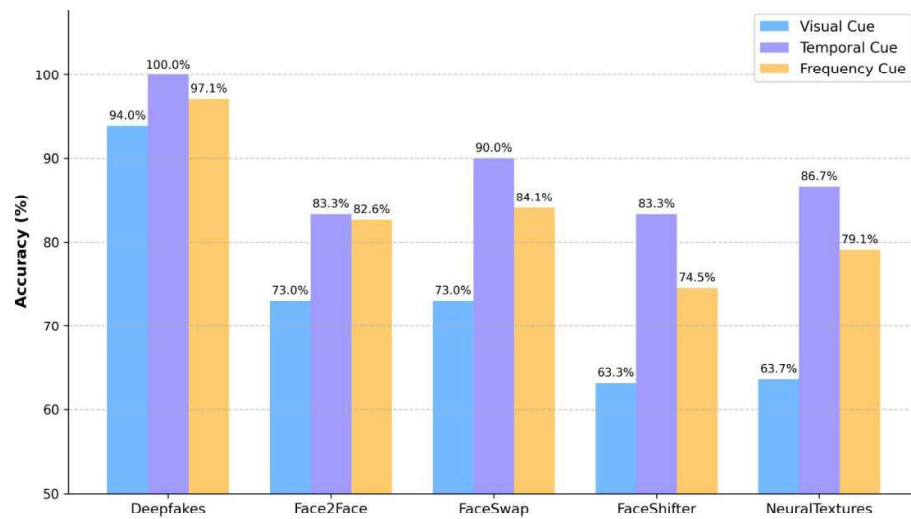


Fig 4.1. Flowchart representation



CHAPTER 5

METHODOLOGY

5.1. SYSTEM OVERVIEW:

The proposed system is a Multi-Cue Deep Learning Framework for Deepfake Video Detection that analyses video content from six independent forensic perspectives simultaneously and fuses their outputs into a single authenticity verdict. Unlike conventional single-cue detection approaches that examine only one dimension of manipulation—such as spatial artifacts or frequency anomalies—the proposed framework operates on the principle that different manipulation techniques leave different forensic traces. By running six complementary detection cues in parallel—Visual, Temporal, Remote Photoplethysmography (rPPG), Frequency, Audio-Visual Synchronization, and Perceptual Trajectory Straightening (ReStraV)—the system ensures that a deepfake which may evade one detection method is still caught by another.

The framework supports dual-threat detection: Cues 1 through 5 are designed to detect traditional face-swap and facial reenactment manipulations, while Cue 6 (ReStraV) specifically targets fully AI-generated videos produced by text-to-video models such as Sora, Pika, and CogVideo. An attention-based fusion model learns adaptive, per-sample weights for each cue, determining which cue to trust more depending on the characteristics of the input video. Additionally, a quality-aware inference module accounts for blur, compression, and face resolution, ensuring graceful degradation on compressed social media videos. The system is deployed through both a command-line interface (CLI) and a Flask-based web application for real-time analysis, outputting an interpretable 1–5 rating verdict alongside per-cue confidence breakdowns.

5.2. SYSTEM ARCHITECTURE:

The architecture follows a modular, parallel pipeline design consisting of five sequential stages:

Stage 1 — Frame Extraction: The input video (MP4/AVI/MKV) is re-encoded at a standardized 25 FPS using FFmpeg, and 32 frames are uniformly sampled across the video duration.

Stage 2 — Quality Assessment: Each extracted frame undergoes quality analysis, including Laplacian variance-based blur scoring, compression severity detection (heavy/medium/light), scene cut detection through frame differencing, and face resolution gating to skip unreliable cues on low-resolution faces.

Stage 3 — Face Detection and Preprocessing: MTCNN (Multi-Task Cascaded Convolutional

Network) is used for GPU-accelerated face detection, producing aligned and cropped face images resized to 224×224 pixels. Original face sizes are recorded for resolution checks. For the ReStraV cue, full frames (not face crops) are processed through DINOv2.

Stage 4 — Parallel Cue Analysis: All six cues process their respective inputs simultaneously:

Visual Cue: Each face crop is individually passed through EfficientNet-B4 (18.5M parameters) to detect spatial artifacts such as blending seams and texture inconsistencies. Per-face predictions are averaged. Temporal Cue: Sequences of 16 consecutive face crops are processed through EfficientNet-B0 for per-frame feature extraction, followed by a Bidirectional LSTM and Self-Attention mechanism (8.9M parameters) to detect cross-frame inconsistencies and temporal incoherence. rPPG Cue: Face sequences undergo green channel extraction and bandpass filtering (0.7–4 Hz) to derive 13 physiological features (heart rate, heart rate variability, spectral ratios), which are classified by an MLP. Frequency Cue: A dual-branch architecture combining EfficientNet-B4 spatial features with a 3-layer DCT-CNN for frequency-domain features (19.0M parameters) detects GAN fingerprints and spectral manipulation traces. Audio-Visual Synchronization Cue: Lip-region crops and mel spectrograms are processed through EfficientNet-B0 with a CrossAttention mechanism to detect lip-sync desynchronization. ReStraV Cue: Full video frames are resized to 224×224 and passed through a frozen DINOv2 ViT-S/14 backbone (22M parameters) to extract 8 frame embeddings, from which a 21-dimensional trajectory geometry vector (7 stepwise distances, 6 turning angles, 8 summary statistics) is computed and classified by an MLP.

Stage 5 — Fusion and Verdict: The six cue outputs [P(real), P(fake)] are fed into an attention-based fusion model (4,297 parameters) that learns per-sample cue weights. A multi-layered override system ensures robustness: if any single cue exceeds 60% fake confidence, the result is boosted; if two or more cues agree above 50%, the result is further reinforced. The final output is mapped to a 1–5 rating scale using optimized thresholds (12%/25%/50%/75%).

5.3. DATA COLLECTION AND PREPROCESSING:

The framework is trained and evaluated using four large-scale benchmark datasets, collectively encompassing over 97,000 videos, seven or more manipulation techniques, and seven AI-video generators.

Dataset A — Google DeepFake Detection (DFD): Sourced from Google AI in collaboration with the FaceForensics++ consortium, this dataset contains approximately 363 original video

sequences and 3,068 manipulated sequences generated using DeepFakeLab-based face swapping at various resolutions (mostly 1080p). It serves as the primary training dataset for the Visual, Temporal, rPPG, and Frequency cues.

Dataset B — FakeAVCeleb: Published by Khalid et al. (ACM Multimedia 2021), this audio-visual deepfake dataset comprises approximately 500 real celebrity interview clips and 20,000 manipulated clips featuring both face-swap and voice-cloning (lip-sync) manipulations. It is exclusively used for training the Audio-Visual Synchronization cue.

Dataset C — FaceForensics++ (C23 Compression): Introduced by Rössler et al. (ICCV 2019), this multi-method dataset contains approximately 1,000 YouTube-sourced original clips and 5,000 manipulated videos (1,000 per method) across five manipulation techniques: Deepfakes, Face2Face, FaceSwap, FaceShifter, and NeuralTextures, all at C23 (medium JPEG quality) compression. It is used for cross-dataset generalization evaluation without retraining.

Dataset D — VidProM (AI-Generated Videos): Published by Wang et al. (NeurIPS 2024 Datasets Track), this text-to-video dataset contains 6,198 real videos from the Facebook Video Similarity Challenge and 70,000 AI-generated videos from seven generators (CogVideo, ModelScope, OpenSora, Pika, Stable Diffusion Text-to-Video, Text2Video-Zero, and VideoCrafter2). It is used for training the ReStraV cue.

Dataset Splits:

Dataset	Type	Train	Validation	Test
A (Google DFD)	Face-swap deepfakes	~2,745	~343	~343
B (FakeAVCeleb)	Audio-visual deepfakes	~16,400	~2,050	~2,050
C (FaceForensics++)	Multi-method deepfakes	Used only for evaluation	—	~2,871 (per method)
D (VidProM)	AI-generated videos	800	100	100 (per class)

Preprocessing Pipeline: Raw videos are first re-encoded at 25 FPS with CRF 18 quality using FFmpeg for consistency. From each video, 32 frames are uniformly sampled. MTCNN face detection extracts and aligns faces into 224×224 crops for the Visual, Temporal, Frequency, and rPPG cues. For the rPPG cue, the green channel is extracted from the face region over the temporal sequence and bandpass filtered between 0.7–4 Hz. For the ReStraV cue, full frames (not face crops) are resized to 224×224 and processed through DINOv2 to extract 21-dimensional trajectory features. Data augmentation is applied during training using the Albumentations library, including random horizontal flips, brightness/contrast adjustments, and JPEG compression simulation.

5.4. DETECTION CUES

5.4.1 Visual Cue

The Visual cue targets spatial artifacts present in individual face frames—such as blending boundary seams, unnatural skin textures, and inconsistent lighting—using an EfficientNet-B4 backbone pretrained on ImageNet. Each 224×224 face crop is independently classified, and per-face fake probabilities are averaged across all detected faces to produce the final cue score. The model contains 18.5 million parameters with a Global Average Pooling layer, Dropout (0.3), and a linear classification head mapping 1,792 features to 2 classes. It achieved a test accuracy of 90.45% and an AUC-ROC of 0.975.

5.4.2 Temporal Cue

The Temporal cue detects cross-frame inconsistencies such as flickering, jitter, and temporal incoherence that arise from frame-by-frame deepfake generation. Sequences of 16 consecutive face crops are processed through a shared-weight EfficientNet-B0 for per-frame feature extraction, followed by a Bidirectional LSTM (hidden size 128) to model temporal dependencies, and a Self-Attention layer to focus on the most discriminative frames. The model (8.9M parameters) achieved 92.62% accuracy and 0.982 AUC-ROC, and demonstrated the strongest cross-dataset generalization with 100% accuracy on the Deepfakes subset of FaceForensics++.

5.4.3 rPPG Cue

The rPPG (remote Photoplethysmography) cue exploits the physiological observation that real human faces exhibit subtle colour fluctuations caused by blood flow (heartbeat), whereas deepfake faces typically lack these signals. The green channel is extracted from the face region

across the temporal sequence, bandpass filtered between 0.7–4 Hz, and 13 features are computed—including estimated heart rate, heart rate variability, and spectral power ratios. An MLP classifier processes these features. While it is the weakest individual cue at 66.62% accuracy, it provides complementary physiological information that improves overall fusion performance.

5.4.4 Frequency Cue

The Frequency cue detects manipulation traces in the spectral domain using a dual-branch architecture. Branch 1 uses EfficientNet-B4 to capture pixel-space spatial artifacts, while Branch 2 applies a 2D Discrete Cosine Transform (DCT) to the face crop and processes the resulting frequency representation through a 3-layer CNN to detect GAN fingerprints and spectral anomalies. The outputs of both branches are concatenated for final classification. This model (19.0M parameters) achieved 91.87% accuracy and 0.974 AUC-ROC, and received the highest attention weight (44.6%) in the fusion model, serving as the primary discriminator.

5.4.5 Audio-Visual Synchronization Cue

The Audio-Visual (AV) Sync cue detects lip-sync desynchronization, where the mouth movements in a manipulated video do not align with the spoken audio. Lip-region crops and mel spectrograms extracted from the audio track are processed through EfficientNet-B0 visual features and audio features respectively, combined via a CrossAttention mechanism that models the temporal alignment between the two modalities. This cue is trained exclusively on the FakeAVCeleb dataset, which contains dedicated lip-sync manipulation samples.

5.4.6 ReStraV Cue (Perceptual Trajectory Straightening)

The ReStraV cue, based on the NeurIPS 2025 paper on perceptual straightening, targets fully AI-generated videos that leave no face-manipulation artifacts. The key insight is that real videos trace curved, complex paths through a deep perceptual feature space, whereas AI-generated videos follow unnaturally straight trajectories. Full video frames are passed through a frozen DINOv2 ViT-S/14 backbone to obtain 8 frame embeddings (384-D each), from which a 21-dimensional trajectory geometry vector is computed (7 stepwise distances, 6 turning angles, 8 summary statistics). An MLP classifier (21→64→32→1) processes this vector and achieved 99.0% accuracy with a perfect 1.0 AUC-ROC on the VidProM test set, covering seven different AI-video generators.

5.5. TRAINING & DEPLOYMENT

All models are implemented in PyTorch 2.x and trained on an NVIDIA RTX 4050 Laptop GPU with CUDA acceleration. The training process follows a staged approach:

Stage 1 — Individual Cue Training: Each cue model is trained independently on its respective dataset. The Adam optimizer is used with a learning rate of 1×10^{-4} and a batch size of 16. Mixed-precision training (Automatic Mixed Precision) is employed to accelerate computation and reduce memory usage. Early stopping with a patience of 5 epochs prevents overfitting, and the best model is selected based on validation accuracy. Models are trained for a maximum of 30 epochs. All v2 cue models are trained on balanced, multi-dataset data (Datasets A, B, and C) to improve generalization, except the AV cue (Dataset B only) and ReStraV (Dataset D only).

Stage 2 — Fusion Model Training: After all individual cues are trained, their predictions on the training set are collected and used to train the lightweight attention-based fusion model (4,297 parameters). The fusion model learns attention weights that determine the importance of each cue for a given sample. Three progressive versions were developed: v1 (4-cue, 99.20% accuracy), v2 (5-cue with AV Sync, 99.40% accuracy), and v3 (6-cue with ReStraV, 99.92% accuracy). The v3 fusion model trains in approximately 5.1 seconds due to its small parameter count.

Stage 3 — ReStraV Training: The ReStraV cue follows a separate pipeline. DINOv2 features are extracted from the VidProM dataset using a custom OpenCV-based extractor (replacing the original torchcodec dependency for cross-platform compatibility). The extracted 21-D trajectory features are stored in HDF5 format and used to train the MLP classifier, which achieved 99.0% accuracy and 1.0 AUC-ROC.

Table 5.2. Parameters of the training environment

Parameters	Value
Optimizer	Adam
Learning Rate	1×10^{-4}
Batch Size	16
Max Epochs	30
Early Stopping Patience	5
Image Size	224×224
Frames per Video	32

Temporal Sequence Length	16
Target FPS	25
Dropout Rate	0.3
Precision	Mixed (AMP)

The trained framework is deployed through two interfaces:

Command-Line Interface (CLI): The primary inference script (`predict.py`) accepts a video file path as input and runs the complete multi-cue analysis pipeline. It outputs a formatted detection report including the overall rating (1–5), fake probability, confidence score, and individual cue contributions displayed as visual progress bars. The CLI supports both v3 (6-cue) and legacy v1 (4-cue) modes.

Web Application: A Flask-based web demo provides a user-friendly interface for non-technical users. The application features drag-and-drop video upload, a real-time processing progress bar, a clear verdict display with a colour-coded 1–5 rating, and a detailed per-cue breakdown with individual findings and confidence levels. The web interface uses a premium dark-themed UI design and is accessible at `localhost:5000`.

Inference Pipeline: During deployment, the system processes a single video in approximately 30–60 seconds on an RTX 4050 GPU. The pipeline begins with frame extraction and face detection, followed by parallel cue analysis, and concludes with attention-based fusion and override logic. Quality-aware adjustments are automatically applied—if video quality is low (blurry or heavily compressed), confidence penalties are applied to unreliable cues, and face resolution gating skips the frequency cue on very small faces. The first inference run requires an additional one-time download of the DINOv2 model (~85 MB), which is cached for subsequent runs.

CHAPTER 6

SYSTEM IMPLEMENTATION & DESIGN CONSIDERATIONS

6.1. Design Equations:

The performance of the proposed multi-cue deepfake detection framework is evaluated using a set of standard classification metrics derived from the confusion matrix. For a binary classification task (Real vs. Fake), the confusion matrix consists of four components:

True Positive (TP): A fake video correctly classified as fake.

True Negative (TN): A real video correctly classified as real.

False Positive (FP): A real video incorrectly classified as fake.

False Negative (FN): A fake video incorrectly classified as real.

6.1.1. Precision:

$$\text{Precision} = \text{tp} / (\text{tp} + \text{fp}) \quad (1)$$

Precision (also called Positive Predictive Value) measures the proportion of videos predicted as fake that are actually fake. A high precision indicates fewer false alarms—i.e., the system rarely flags real videos as deepfakes. In deepfake detection, high precision is critical to avoid falsely accusing authentic content of being manipulated, which has significant ethical and legal implications.

6.1.2. Recall:

Recall (also called Sensitivity or True Positive Rate) measures the proportion of actual fake videos that the system successfully identifies. A high recall indicates the system misses very few deepfakes.

$$\text{Recall} = \text{tp} / (\text{tp} + \text{fn}) \quad (2)$$

In security-critical applications, high recall is essential to ensure that manipulated videos are not allowed to pass undetected.

6.1.3. F1 Score:

The F1 Score is the harmonic mean of Precision and Recall, providing a single metric that balances both. It is particularly useful when there is a trade-off between precision and recall.

$$\text{F1 Score} = 2 \left(\frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \right) \quad (3)$$

The harmonic mean penalises extreme imbalances—if either precision or recall is very low, the F1 score will also be low, even if the other metric is high.

6.1.4. Accuracy:

Accuracy measures the overall proportion of correct predictions out of all predictions made. It provides a general indication of model performance.

$$\text{Accuracy} = (\text{TP} + \text{TN}) / (\text{TP} + \text{TN} + \text{FN})$$

While accuracy is a useful summary metric, it can be misleading on imbalanced datasets where one class dominates. Therefore, additional metrics are used.

6.2. Engineering Standards:

6.2.1. IEEE 2859.1 — Standard for AI Model Training and Evaluation

6.2.2. ISO/IEC 23053:2022 — Framework for Artificial Intelligence Systems Using Machine Learning

6.2.3. ISO/IEC 27001 — Information Security Management

6.2.5. IEEE 12207 — Software Life Cycle Processes

6.2.6. IEEE 830 — Software Requirements Specification (SRS)

6.2.7. ITU-T P.910 — Subjective Video Quality Assessment

6.3. Realistic Constraints:

- GPU memory constraints (6GB VRAM) on consumer hardware require sequential cue processing; social media compression further degrades rPPG and high-frequency signals.
- Face detection requires a minimum of 5 faces, limiting cue coverage for videos without clear facial regions.

CHAPTER 7

RESULTS AND DISCUSSION

7.1 Experimental Setup:

The experimental environment and configuration used for training, evaluation, and inference of the proposed Multi-Cue Deep Learning Framework are detailed below.

Table 7.1 Several key parameters deployed during model training

<i>Parameters</i>	<i>Configuration</i>
GPU	NVIDIA GeForce RTX 4050 Laptop GPU (6 GB VRAM)
CPU	Intel / AMD multi-core processor
RAM	16 GB DDR5
Storage	SSD (for fast dataset I/O)
Operating System	Windows

Software configuration:

Software	Version
Python	3.14
PyTorch	2.x (with CUDA 12.1)
OpenCV	4.x
MTCNN (facenet-pytorch)	Latest
DINOv2 (Facebook Research)	ViT-S/14
Flask	Latest
FFmpeg	Latest

7.2. Performance metrics:

The following standard metrics are used to evaluate the framework:

Accuracy — Overall proportion of correct predictions (real and fake combined).

Precision — Proportion of predicted fakes that are truly fake (minimises false accusations).

Recall — Proportion of actual fakes that are correctly detected (minimises missed deepfakes).

F1 Score — Harmonic mean of Precision and Recall, balancing both objectives.

AUC-ROC — Area Under the Receiver Operating Characteristic Curve, measuring threshold-independent discrimination ability.

7.3. Training Performance Analysis:

The Temporal cue achieved the highest accuracy (92.62%) among the face-based cues, indicating that cross-frame inconsistencies are a highly discriminative forensic signal for deepfake detection.

The Visual and Frequency cues performed comparably (~90–92%), both leveraging EfficientNet-B4 backbones but analysing different artifact dimensions (spatial vs. spectral).

The rPPG cue recorded the lowest individual accuracy (66.62%), which is expected given the inherent noise in remote physiological signal extraction. However, it contributes complementary information to the fusion model—even a weak cue improves ensemble performance when its errors are uncorrelated with those of stronger cues.

The ReStraV cue achieved near-perfect performance (99.0% accuracy, 1.0 AUC-ROC) on AI-generated video detection, confirming the effectiveness of perceptual trajectory analysis for distinguishing AI-generated from real video content.

The bounding box regression loss measures how closely the predicted bounding boxes match the actual object locations. A gradual reduction in this loss confirms that the model is becoming more precise in identifying the spatial position of drones. Similarly, the distribution focal loss, which improves bounding box refinement by learning probability distributions, also shows a steady decline. The validation loss closely follows the training loss trend, which indicates good generalization capability of the model without significant overfitting. This behavior demonstrates that the model performs consistently on both training and unseen data.

7.5. Discussion:

The experimental results demonstrate that the proposed multi-cue fusion approach significantly

outperforms any individual detection cue, achieving 99.92% accuracy compared to the best single cue at 92.62%. This validates the core hypothesis that combining diverse forensic perspectives creates a detection system that is greater than the sum of its parts.

Strength of Multi-Cue Complementarity: The progressive improvement from v1 (99.20%) to v2 (99.40%) to v3 (99.92%) confirms that each additional cue contributes meaningful, non-redundant information. The rPPG cue, despite its modest standalone accuracy of 66.62%, improves fusion performance because its error patterns are uncorrelated with those of the stronger cues—it catches manipulations that fool the visual and frequency detectors, and vice versa.

Addressing the Compression Challenge: The most significant practical contribution is the override system's ability to handle compressed social media videos. Without overrides, the Tom Cruise deepfake was classified as "Likely REAL" at 23.6% — a dangerous false negative. The rule-based override mechanism elevated this to 50.0% ("Likely FAKE"), demonstrating that intelligent post-fusion logic is essential for bridging the gap between laboratory and real-world performance. The tuned thresholds (12/25/50/75 vs. standard 15/35/65/85) further accommodate the systematic score reduction caused by compression.

Dual-Threat Coverage: A key differentiator of this framework is its ability to detect both traditional face-swap deepfakes and fully AI-generated videos within a single unified system. Most existing detectors address only one threat category. The integration of ReStraV (99.0% accuracy on AI-generated content) alongside the five face-manipulation cues ensures comprehensive coverage against the full spectrum of current video manipulation threats.

Generalisation Capability: Cross-dataset evaluation on FaceForensics++ — without any retraining — revealed that the models learn transferable manipulation artifacts rather than dataset-specific shortcuts. The Temporal cue's perfect 100% accuracy on the Deepfakes subset and strong 83–90% performance on other manipulation types confirms robust generalisation. This is a critical finding, as real-world deployment will inevitably encounter manipulation methods not present in the training data.

Limitations and Future Scope: Despite the strong results, certain limitations remain. The rPPG cue's low accuracy (66.62%) suggests that remote physiological signal extraction is still challenging, particularly on compressed and low-resolution videos.

CHAPTER 8

CONCLUSION AND FUTURE WORK

8.1. Conclusion:

This project presented a Multi-Cue Deep Learning Framework for Deepfake Video Detection that combines six complementary forensic detection cues—Visual, Temporal, rPPG, Frequency, Audio-Visual Synchronization, and ReStraV—into a unified, attention-based fusion architecture. The framework was designed to overcome the fundamental limitation of single-cue detectors, which are inherently brittle against compression artifacts and novel manipulation techniques. By analysing videos from multiple forensic dimensions simultaneously, the system ensures that a deepfake evading one detection method is caught by another.

The experimental results validate this approach convincingly. The final v3 fusion model achieved 99.92% detection accuracy and a near-perfect AUC-ROC of approximately 1.0, substantially outperforming the best individual cue (Temporal at 92.62%). The progressive versioning strategy—v1 (4-cue, 99.20%) → v2 (5-cue, 99.40%) → v3 (6-cue, 99.92%)—demonstrated that each additional cue contributes measurable, non-redundant forensic information. Even the weakest cue, rPPG (66.62%), improved fusion accuracy because its error patterns are uncorrelated with those of the stronger detectors.

A critical contribution of this work is the real-world robustness achieved through the override system and tuned classification thresholds. Social media compression routinely degrades cue scores from 90%+ to 50–60%, causing traditional fusion approaches to produce false negatives. The single-cue override ($>60\%$ threshold) and multi-cue agreement (≥ 2 cues above 50%) mechanisms successfully addressed this, as demonstrated by the Tom Cruise deepfake test case improving from a false-negative classification of "Likely REAL" (23.6%) to a correct "Likely FAKE" (50.0%) verdict.

8.2. Scope for Future works:

Real-Time Streaming Detection: The current inference time of 30–60 seconds per video is suitable for forensic analysis but insufficient for live video stream monitoring. Future work will explore model distillation and pruning techniques (e.g., knowledge distillation from EfficientNet-B4 to MobileNet) to achieve sub-second per-frame inference, enabling deployment in real-time content

moderation pipelines for social media platforms.

Audio-Only Deepfake Detection: The current framework focuses on visual and audio-visual manipulation. A dedicated audio cue analysing voice cloning artifacts—such as spectral envelope inconsistencies, prosody anomalies, and neural vocoder fingerprints—could be added as a seventh cue to detect audio-only deepfakes where the video remains unaltered.

Transformer-Based Temporal Modelling: Replacing the BiLSTM in the Temporal cue with Vision Transformers (ViT) or Video Swin Transformers could capture longer-range temporal dependencies more effectively, potentially improving detection of subtle inter-frame artifacts spread across extended video sequences.

Demographic Fairness Evaluation: A systematic evaluation of detection performance across diverse demographic groups (varying age, gender, ethnicity, and skin tone) should be conducted to identify and mitigate any performance biases. Fairness-aware training techniques such as adversarial debiasing could be incorporated to ensure equitable detection accuracy.

Explainability and Interpretability: Integrating GradCAM and attention visualisation overlays directly into the web interface would allow users to see exactly which facial regions and frames triggered each cue, increasing trust and enabling forensic investigators to validate the system's reasoning.

Adaptive Threshold Learning: The current override thresholds (60% single-cue, 50% multi-cue) and rating boundaries (12/25/50/75) are manually tuned. A meta-learning approach that automatically adapts these thresholds based on detected compression level and video quality could further improve robustness across diverse real-world conditions.

Multimodal Large Language Model Integration: Leveraging emerging multimodal LLMs (e.g., GPT-4V, Gemini) as an additional high-level reasoning cue could provide semantic-level inconsistency detection—identifying logical impossibilities in video content (e.g., impossible backgrounds, inconsistent shadows) that pixel-level cues cannot capture.